Design Verification Plan & Report

VBF-T8R1NOFORCE-DVPR-01 | case t8_r1_noforce | 2026-08-25 | virtual design verification (simulation-based)

Executed verification items

#	Item	Method / condition	Result	Criterion	Verdict
1	Gravimetric energy density	calc-energy contract caliber	497.968 Wh/kg	>= 446.18 Wh/kg	PASS
2	5C capacity retention	DFN 5C vs 1C, same params	0.9896	>= 0.90	PASS
3	Cell mass	calc-energy layer mass x area	38.99 g	<= 40 g	PASS
4	Plating under fast charge	4C at 45 C, plating module	anode min +0.00635 V -> plated=false	plated = false	PASS (margin +6.3 mV - flagged)
5	T_max fast charge	same run, lumped thermal	326.81 K	<= 333.15 K	PASS
6	T_max 5C discharge	DFN 5C discharge, 25 C	313.64 K	<= 333.15 K	PASS
7	1C discharge capacity	DFN 1C discharge, 25 C	5.2725 Ah	no entry-0 threshold (feeds 1-2)	PASS (informational)

Result: 7 / 7 executed items PASS. All five entry-0 stage2/stage3 criteria verified green on final design (R6-V9); verification trail: log.jsonl round 6 propose -> evaluate (checked = 5, mechanical).

Uncovered conditions (honest N/A list)

Condition	Reason not executed	Materiality
Cycle life / long-term aging	not a task criterion; not simulated (no fabricated values)	medium - endurance drones accumulate cycles
Nail penetration	requires physical experiment	low for pouch drone pack
Crush / drop / vibration	requires physical experiment	medium - airframe vibration real; recommend qualification
Thermal runaway abuse	protocol exists but out of task scope; not run	low (4C exam covers task thermal red line)
Low-temperature retention (-20 C)	not a task criterion	medium - altitude operation is cold
Self-discharge / storage	not simulated	low (single-flight-cycle pattern)
DCR pulse measurement	calc-energy DCR is formulation estimate, not pulse test	low (power density far exceeds demand)

Verification conclusion

The R6-V9 design satisfies all entry-0 acceptance criteria. Two margins deserve engineering attention before physical prototyping: (a) plating margin on 4C charge is +6.3 mV - thin; recommended mitigation is a 45 C charge-temperature floor and/or 3C charge in cold conditions (see DFMEA); (b) the 10 um separator and 6-8 um current collectors are aggressive and must be validated for pouch winding/lasering capability. These are recorded as risks, not criterion failures - the criteria themselves are all met.